

How an Analyst Approaches This Problem

The reasoning behind the Building Permits Live Tracker — the questions asked, the decisions made, and what the analysis actually found.

Note on the numbers in this document: the notebook currently runs in simulated-data mode (`USE_LIVE_API = False`) so it can be tested without waiting on a live pull. The figures below come from that simulated run and are structurally realistic, but are not real StatCan releases. Once you set `USE_LIVE_API = True` and re-run, replace every number here with the live output — the reasoning and structure of this document stay exactly the same.

Step 1 — Start with the business question, not the data

A common beginner mistake is opening a dataset and asking "what can I find in here?" An analyst starts from the other direction: what decision is someone trying to make, and what would they need to know to make it? For this project, the framing question was:

"Is construction activity in Canada accelerating or slowing, and does that story differ by region?"

That single question is what determined everything downstream — which table to pull, which regions to keep, and which analysis techniques were worth the effort. Framing it this way also mirrors a real workplace ask: nobody requests "building permit data" for its own sake, they're deciding where to invest, staff, or price something.

Step 2 — Choose the data source deliberately

Statistics Canada's Building Permits table (34-10-0066-01) was chosen over alternatives (CMHC housing starts, provincial construction associations) for three reasons worth being able to defend in an interview:

- It's a **leading indicator** — permits are issued before construction starts, so the data signals what's coming, not just what already happened
- It's **free and fully public** with no signup friction, which matters for a reproducible pipeline anyone can rerun
- It's released **monthly and by region**, which supports both a national narrative and a regional comparison in the same table

Step 3 — Decide what to keep and what to cut

The live table includes every province, every census metro area, every structure and work-type combination — hundreds of thousands of rows. Pulling all of it isn't rigor, it's noise. The decision was to keep:

- **National total** — the headline number any reader wants first
- **Four provinces (Alberta, Ontario, British Columbia, Quebec)** — the largest construction markets by volume, chosen because they're where a regional story is most likely to be meaningful, and because they map to where the job search is actually targeting
- **Residential vs. non-residential split** — because a residential slowdown and a non-residential slowdown imply different things (housing demand vs. commercial investment) and shouldn't be blended into one number

Cutting census metro areas and minor work-type categories was a deliberate scope decision, not an oversight — a dashboard trying to show everything ends up showing nothing clearly.

Step 4 — Match the analysis technique to the question, not the other way around

Three different techniques were used, each chosen for what it specifically answers — not stacked on top of each other for the sake of using more tools:

"How does this compare to last year?" → year-over-year % change

The most basic question a stakeholder asks first. In the current run, the national total for the most recent month sits about **3.0% above the same month last year** — a modest year-over-year gain, not a dramatic shift in either direction.

"Is that a real trend, or just the season?" → seasonal decomposition

Construction is seasonal by nature — permits dip every winter and rise every spring regardless of the underlying economic trend. Without separating that out, a January dip could be misread as a slowdown. Decomposing the series into trend, seasonal, and residual components isolates the genuine underlying direction from the pattern that repeats every year regardless.

"Where is this likely headed next quarter?" → linear trend forecast

Once the underlying trend is isolated, a simple linear regression on the last 24 months gives a defensible 3-month forward estimate. In the current run, the trend direction comes out **upward**, projecting roughly 13,700–13,900 in national permit value over the next three months — a short-term, directional estimate deliberately kept simple enough to explain in one sentence, rather than a black-box model nobody in the room could evaluate.

Step 5 — What the regional comparison actually found

Comparing each province's most recent 3-month average against the 3 months before it (a simple momentum check, not a formal statistical test) produced a clear ranking in the current run:

Region	Momentum (recent 3mo vs. prior 3mo)
British Columbia	+5.3% — fastest-growing
Quebec	+3.8%
Alberta	+1.7%
Ontario	+0.2% — roughly flat

The interpretation an analyst would offer here isn't just the ranking — it's what it implies: British Columbia and Quebec are showing the strongest near-term momentum, while Ontario, despite being the largest market by volume, is essentially flat. That's a more useful finding than the national number alone, because it tells a reader *where* the national growth is concentrated rather than assuming it's evenly spread.

Step 6 — Name the limitations honestly

A finished-looking dashboard can create false confidence if its limitations aren't stated. Worth being upfront about, both here and in an interview:

- The 3-month momentum comparison is a simple average-vs-average check, not a statistically tested trend — it can be noisy for a single volatile month
- The linear forecast assumes the recent trend continues; it will not anticipate a policy change, interest rate move, or other structural shift
- StatCan permit data reflects *intent* to build, not completed construction — actual housing starts can lag or fall short of permitted volume

Step 7 — The takeaway, stated the way you'd say it out loud

"National building permit activity is up modestly year-over-year and trending upward over the next quarter, but that growth isn't evenly distributed — British Columbia and Quebec are accelerating while Ontario, the largest market, is essentially flat. If I were prioritizing where to focus analysis or outreach next, I'd look at what's driving BC's momentum specifically, since it's the strongest signal in the data."

That closing line is deliberate: an analyst's output isn't just a chart, it's a recommendation for what to look at next. Ending on "here's what I'd do with this" is what separates a reporting exercise from an analytical one — and it's the version of this project worth having ready to say in an interview.

For the technical how-it-works (API mechanics, code walkthrough), see [StatCan_Tracker_Explainer.pdf](#). This document covers the reasoning; that one covers the implementation.